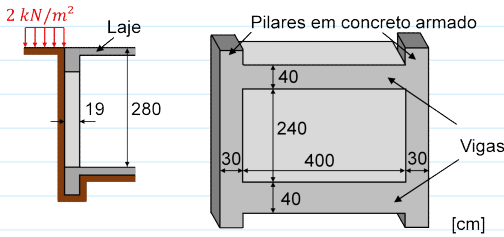


Muro de Arrimo Integrado

▪ Dimensione o muro de arrimo

⇒ Blocos de concreto: $f_{bk} = 8 \text{ MPa}$

⇒ Solo: $K_a = 0,4$ e $\gamma = 20 \text{ kN/m}^3$



Relação de cálculo

$$\min\left(\frac{30}{2}; \frac{19}{2}\right) + 400 + \min\left(\frac{30}{2}; \frac{19}{2}\right) = 419 \text{ cm}$$

$$\min\left(\frac{40}{2}; \frac{19}{2}\right) + 240 + \min\left(\frac{40}{2}; \frac{19}{2}\right) = 259 \text{ cm}$$

Emprego:

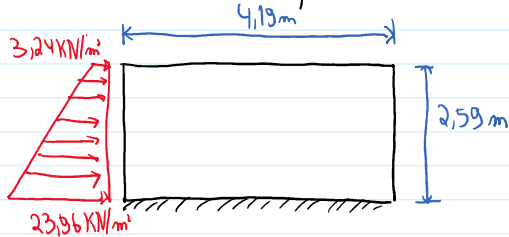
Solo: $\sigma_v = 2 + 20 \cdot 0,305 = 8,1 \text{ kN/m}^2$

Barras: $\sigma_v = 2 + 20 \cdot 2,895 = 59,9 \text{ kN/m}^2$

$\sigma_{ha} = 0,4 \cdot 8,1 = 3,24 \text{ kN/m}^2$

$\sigma_{ha} = 0,4 \cdot 59,9 = 23,96 \text{ kN/m}^2$

Modelo estrutural com ações características



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Tabela de Dimensionamento: Barras 2B + Barras 12

Barras 2B:

$l_y = 4,19 \text{ m}$

$l_x = 2,59 \text{ m}$

$\lambda = l_y/l_x = 4,19/2,59 = 1,618$

$M_x = 5,472$

$M'_x = 11,403$

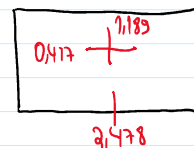
$M_y = 1,921$

$m_x = 5,472 \cdot 0,2173 = 1,189 \text{ kN/m/m}$

$m'_x = 11,403 \cdot 0,2173 = 2,478 \text{ kN/m/m}$

$m_y = 1,921 \cdot 0,2173 = 0,417 \text{ kN/m/m}$

$\frac{p l^2}{100} = \frac{3,24 \cdot 2,59^2}{100} = 0,2173 \text{ kN/m/m}$



Barras 12:

$l_a = 2,59 \text{ m}$

$l_b = 4,19 \text{ m}$

$\gamma = l_a/l_b = 2,59/4,19 = 0,618$

$p = 23,96 - 3,24 = 20,72 \text{ kN/m}^2$

$M_x = 2,610$

$M'_x = 6,232$

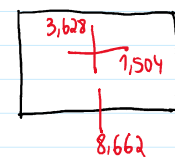
$M_y = 1,082$

$m_x = 2,610 \cdot 1,139 = 3,628 \text{ kN/m/m}$

$m'_x = 6,232 \cdot 1,139 = 8,662 \text{ kN/m/m}$

$m_y = 1,082 \cdot 1,139 = 1,504 \text{ kN/m/m}$

$\frac{p l^2}{100} = \frac{20,72 \cdot 2,59^2}{100} = 1,39 \text{ kN/m/m}$



Solutores característicos

$m_x = 1,189 + 3,628 = 4,817 \text{ kN/m/m}$

$m'_x = 2,478 + 8,662 = 11,14 \text{ kN/m/m}$

$m_y = 0,417 + 1,504 = 1,921 \text{ kN/m/m}$

$$m'_x = 2,478 + 8,662 = 11,14 \text{ KN/m/m}$$

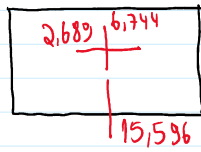
$$m_y = 0,417 + 1,504 = 1,921 \text{ KN/m/m}$$

Solutorio de colunas

$$M_{dx} = 1,4 \cdot 4,817 = 6,744 \text{ KN/m/m}$$

$$M_{dxe} = 1,4 \cdot 11,14 = 15,596 \text{ KN/m/m}$$

$$M_{dy} = 1,4 \cdot 1,921 = 2,689 \text{ KN/m/m}$$



Dimensionamento à flexão

Flexão vertical

Afastando armaduras excêntricas, simétricas

$$M_{sd} = 1560 \text{ KN cm/m}$$

$$f_{pk, lq} = 0,6 / 0,54 = 1,111 \text{ KN/cm}^2$$

$$K = 1,0$$



$$d = 19 - 6 = 13 \text{ cm}$$

$$b_m = 20 \text{ cm} \quad A_s = 0,64 \text{ cm}^2$$

$$b_m = 40 \text{ cm} \quad A_s = 1,28 \text{ cm}^2$$

$$b_m = 60 \text{ cm} \quad A_s = 1,92 \text{ cm}^2$$

$$\Phi 10 \text{ c/} 20 \text{ cm}$$

$$\Phi 16 \text{ c/} 40 \text{ cm}$$

$$\Phi 16 \text{ c/} 60 \text{ cm}$$

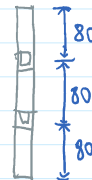
Flexão horizontal

$$M_{sd} = 269 \text{ KN cm/m}$$

Afastando armadura centrada $d = 9,5 \text{ cm}$

$$f_{pk, lq} = 1,111 \text{ KN/cm}^2$$

$$K = 0,5$$



$$b_m = 80 \text{ cm} \quad A_s = 1,14 \text{ cm}^2 \quad \Phi 12,5 \text{ c/} 80 \text{ cm}$$

